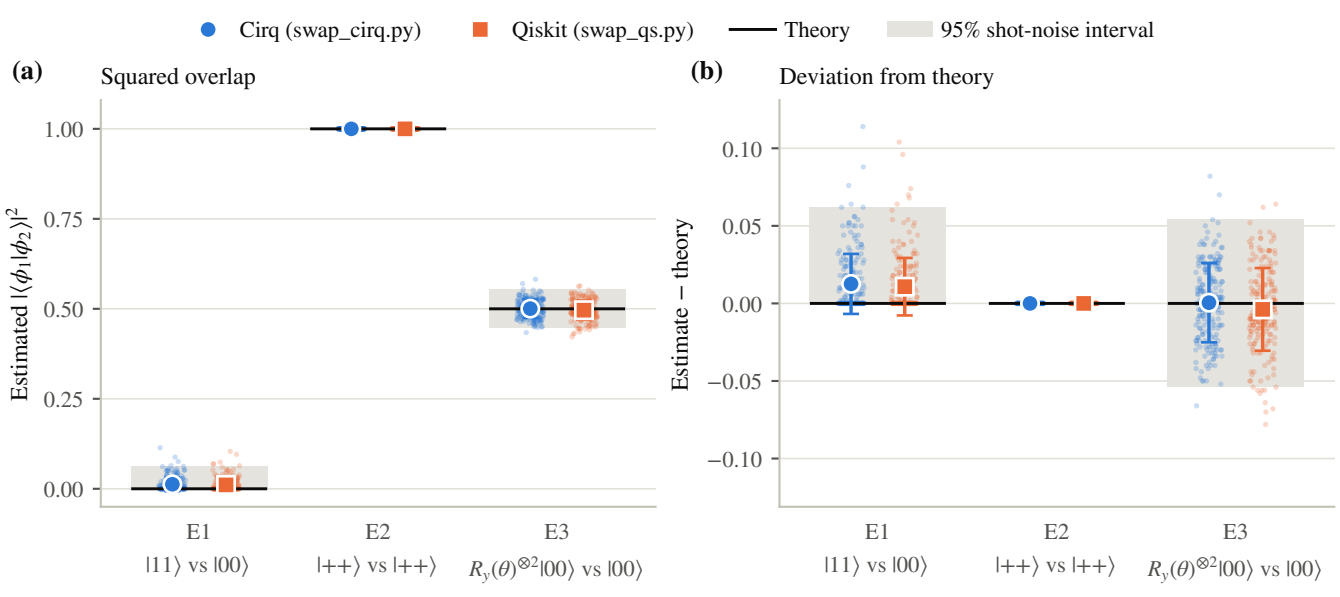
Cirq (swap_cirq.py)
Qiskit (swap_qs.py)
Theory
95% shot-noise interval
(a)
Squared overlap
Estimated $|\langle\phi_1|\phi_2\rangle|^2$
1.00
0.75
0.50
0.25
0.00
E1
$|11\rangle$ vs $|00\rangle$
E2
$|{+}{+}\rangle$ vs $|{+}{+}\rangle$
E3
$R_y(\theta)^{\otimes 2}|00\rangle$ vs $|00\rangle$
(b)
Deviation from theory
Estimate − theory
0.10
0.05
0.00
−0.05
−0.10
E1
$|11\rangle$ vs $|00\rangle$
E2
$|{+}{+}\rangle$ vs $|{+}{+}\rangle$
E3
$R_y(\theta)^{\otimes 2}|00\rangle$ vs $|00\rangle$